

Hierarchical Game-Based Optimal Security Control of Vehicular Platoon Systems Against FDI Attacks

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Abstract—This study investigates the optimal security control (OSC) problem for Intelligent Connected Vehicle Platoon Systems (ICVPS) against malicious false data injection (FDI) attacks. A hierarchical game-based control framework is proposed, which integrates differential graph games and zero-sum dynamic adversarial games. This framework is designed to guarantee both global stability and local security performance of the ICVPS. In the outer layer addressing network communication, a differential graph game model is established, treating each vehicle as an independent agent. This enables the entire ICVPS to achieve Nash equilibrium, thereby ensuring optimal global platoon stability and cooperative control precision. In the inner layer focusing on vehicle-level control, a zero-sum dynamic adversarial game model is constructed. Here, the designed optimal security controller and state information-based malicious FDI attacks act as opposing players, ensuring optimal vehicle-level security performance under dual-channel FDI attacks. Furthermore, an improved offline-learning residual neural network (OLRNN) integrated with a cosine annealing mechanism is designed to solve the optimal security controller. This approach significantly enhances training efficiency and effectively prevents the OLRNN from falling into local optima. Finally, simulation experiments conducted on the CARLA simulator verify that the proposed algorithm can effectively mitigate vehicle security issues caused by malicious FDI attacks with minimal control cost. Additionally, ablation studies confirm the superiority of the proposed OLRNN model.

Index Terms—Vehicular platoon systems, false data injection attacks, hierarchical game security control, offline learning.

I. INTRODUCTION

WITH the advancement of artificial intelligence and autonomous driving technologies, intelligent vehicles and vehicle-to-everything (V2X) technology have rapidly become core components of modern transportation systems [1]. The ICVPS are considered as a key V2X application scenario, where

a coordinated group of vehicles maintains preset distances and velocities, achieving cooperative movement while ensuring security features such as collision avoidance, thereby forming an orderly platoon [2], [3]. By integrating with communication networks, ICVPS can reduce the computational demands on individual vehicles and enhance road traffic efficiency. To achieve efficient management and control of vehicles within the ICVPS—while meeting modularity and scalability requirements and ensuring compatibility with industry-standard communication protocols—a layered control and communication architecture [4] has become the mainstream framework for current development. Specifically, considering the high real-time demands for executing control commands, it is a scientifically sound and rational choice to delegate data processing and computation tasks to high-performance cloud servers, while the vehicles themselves are responsible solely for executing the control commands. This strategy effectively alleviates the computational burden on the vehicles but simultaneously raises higher demands for network security.

Due to the openness of wireless networks and complex dynamic traffic environments, ICVPS are highly susceptible to severe network attacks during operation. These attacks can generally be categorized into Denial-of-Service (DoS) attacks, spoofing attacks, and FDI attacks, based on their modes of occurrence and effects [5], [6]. Among these, DoS and spoofing attacks are more common, and many studies have effectively mitigated their practical impact [7], [8], [9], [10]. In contrast, FDI attacks, which have a higher technical threshold, have gradually become one of the emerging research hotspots [11]. For cyber-physical systems (CPS), methods involving distributed pre-selectors and observers have been proposed to ensure secure state estimation under sensor attacks and maintain dynamic surface control [12].

In ICVPS, FDI attacks can originate from external adversaries or compromised internal components. External attackers can exploit vulnerabilities in Vehicle-to-Vehicle (V2V) or Vehicle-to-Infrastructure (V2I) communication links (e.g., man-in-the-middle attacks, message injection) to insert falsified state or control data, or use external devices (such as laser emitters or radio-frequency jammers) to spoof sensors, thereby inferring or disrupting vehicle states. Internal attacks may occur on on-board sensors, such as cameras, LiDAR, millimeter-wave radar, and ultrasonic sensors, which could be maliciously tampered with or fed falsified signals [13]. Furthermore, control commands and perception data are transmitted between the network

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communication layer and the vehicle control layer. FDI attacks pose a significant threat to the reliability, availability, and integrity of system communication data by deliberately tampering with measurement or communication data, leading the controller to operate based on incorrect information. For instance, injecting falsified position, velocity, or acceleration data can distort the estimation of inter-vehicle distance, resulting in unsafe spacing policies. Similarly, falsifying control commands may directly degrade the system's stability, safety, and efficiency. Therefore, developing effective strategies to mitigate or eliminate the impact of FDI attacks on actuators and sensors is crucial for the stable and secure operation of ICVPS.

Current research often focuses on the identification and defense against attacks. Existing methods include combined learning and model-based state estimation to mitigate FDI attacks under disturbances [14], event-triggered mechanisms reducing communication frequency to ensure secure platooning [15], and adaptive control schemes employing Nussbaum functions for CPS under sensor attacks [16]. However, these methods neglect the dynamic interplay between attacks and system performance, often resulting in significant performance degradation or high defense costs [17]. In contrast to traditional FDI attacks that use fixed or bounded random functions, malicious FDI attacks leverage real-time state information, increasing their stealth and complexity. This intelligence makes traditional defense methods ineffective [18]. A Nash equilibrium seeking algorithm is introduced in [19] to analyze the adversarial evolutionary mechanism of attacks and to achieve security control in multi-agent systems. This provides a new perspective for addressing security control while considering system performance. In this framework, the attack and system performance are in a direct adversarial relationship in the game—when the attacker benefits from the game, the system performance inevitably suffers. Therefore, when malicious FDI attacks occur, how to achieve optimal system performance through game-based control becomes a new direction of research.

In ICVPS, system performance encompasses global control accuracy (platoon coordination) and the robustness and stability of vehicles. These are typically considered as global performance and local performance. A differential game theory approach is applied in [20] to address the distributed optimal fault estimation and fault-tolerant control problems for interconnected systems with faults and their diagnostic observer interactions, providing a reference for achieving both global and local optimality. However, practical networked ICVPS involve heterogeneous vehicles with distinct physical properties and varying control demands (e.g., platoon positions, lengths, and weights) [21], complicating their interactions and coordination. Meanwhile, in game theory problems, the Hamilton-Jacobi-Isaacs (HJI) equation exhibits strong nonlinear partial differential equation characteristics, making analytical solutions difficult to obtain [22]. Existing research typically uses adaptive neural networks to approximate these nonlinear issues [23], [24]. An online adaptive learning algorithms have been developed to approximate optimal solutions in multi-agent games [25], thus solving optimal dynamic programming problems. However, the iterative nature of online learning presents significant

challenges for practical applications in ICVPS. If real-world vehicles rely on online reinforcement learning that requires repeated experiments, this could lead to collisions or loss of control. Therefore, a key challenge that remains is how to solve the HJI equation through offline learning, balancing the optimal local performance of individual vehicles with the optimal global performance of the platoon within the framework of game theory.

To address these challenges, this study proposes a hierarchical game-based security control framework for ICVPS. At the network communication layer, a differential graph game theory approach is applied, modeling each vehicle as an independent agent. The goal is to achieve the Nash equilibrium of the entire ICVPS and ensure optimal platoon stability and coordination control accuracy. At the vehicle control layer, a zero-sum game relationship is established between the controller and malicious FDI attacks. Based on the OLRNN, the security controller is developed to effectively suppress the vehicle performance degradation and system instability caused by malicious FDI attacks, while minimizing costs. The main contributions of this study are as follows:

- 1) A novel hierarchical game-based security control approach is proposed to address the ICVPS control challenges under malicious FDI attacks. This approach effectively handles more intelligent and targeted malicious FDI attacks while coordinating the coupling contradiction between the robustness and stability of individual vehicle systems and the global cooperative accuracy of the platoon, enhancing the flexibility and practicality of the control strategy.
- 2) An innovative coupling of differential graph game theory and zero-sum game theory is used to create a “game of games” in ICVPS. The optimal security controller is designed using a zero-sum game at the vehicle control layer, ensuring internal robustness and stability at minimal cost. At the network communication layer, each vehicle is modeled as a participant in a differential graph game, driving the ICVPS towards Nash equilibrium and ensuring optimal platoon performance and cooperative accuracy. This approach ensures both local optimality and global performance optimization, enhancing system security and performance under malicious FDI attacks.
- 3) An OLRNN based on an improved cosine annealing mechanism is proposed for approximating the optimal control analytical solution. By dynamically adjusting the learning rate through estimating the maximum curvature of the loss function, the model avoids getting trapped in local optima, while also reducing training complexity and simplifying experimental application processes.

The remainder of this study is organized as follows: Section II introduces the preliminaries and problem formulation for ICVPS against FDI attacks. Section III presents the hierarchical game-theoretic OSC framework for ICVPS, along with its implementation process. Section IV offers a comprehensive discussion of the simulation results and analysis using the CARLA simulator. Finally, Section V concludes the study.

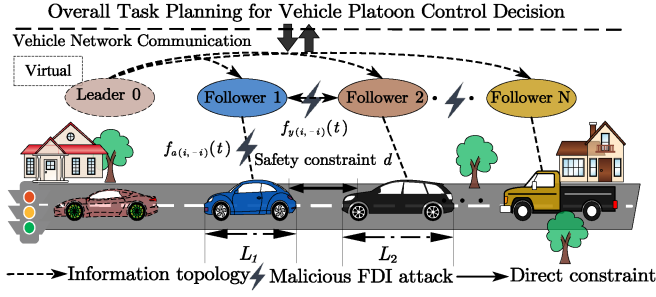


Fig. 1. ICVPS against malicious FDI attacks.

II. PRELIMINARIES AND PROBLEM FORMULATION

The directed graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N} = \{1, 2, \dots, n\}$ denotes the set of nodes corresponding to individual vehicles, and $\mathcal{E}$ denotes the set of edges among the vehicles. The neighbors' set of vehicle i is denoted as N_i .

A. ICVPS Model Against Malicious FDI Attacks

Consider the ICVPS consisting of $n + 1$ intelligent vehicles, where vehicle 0 serves as a virtual leader, and the remaining vehicles are designated as followers. The dynamics of the i -th follower vehicle ($i = 1, 2, \dots, n$) are described as follows:

$$\begin{cases} \dot{p}_i(t) = v_i(t) \\ \dot{v}_i(t) = a_i(t) \\ \dot{a}_i(t) = K_i(t) + \frac{1}{\tau_i} u_i(t) + f_{a(i,-i)}(t) \\ \quad + w_i(t) + \sum_{j \in N_i} \pi_{ij} f_{a(j,-j)}(t) \\ y_i(t) = p_i(t) + f_{y(i,-i)}(t) + \sum_{j \in N_i} \pi_{ij} f_{y(j,-j)}(t) \end{cases} \quad (1)$$

where $p_i(t) \in \mathbb{R}$, $v_i(t) \in \mathbb{R}$, $a_i(t) \in \mathbb{R}$ denote the position, velocity, and acceleration of vehicle i . $u_i(t) \in \mathbb{R}$, $y_i(t) \in \mathbb{R}$, τ_i denote the control input, the system output, and the engine time lag. $K_i(t) \in \mathbb{R}$ accounts for unmodeled dynamic components, and $w_i(t) \in \mathbb{R}$ denotes external disturbances. The ICVPS adopt the communication topology in Fig. 1, where $-i \triangleq \{j : j \in N_i\}$ denotes the set of vehicles physically adjacent to vehicle i .

Moreover, $f_{a(i,-i)}(t) \in \mathbb{R}$ and $f_{y(i,-i)}(t) \in \mathbb{R}$ denote FDI attacks targeting the control input and sensor output channels. These attacks are formed by the combination of direct and coupled attacks on the respective channels, and both are differentiable. The malicious FDI attacks propagating through the vehicular communication network in the ICVPS can be expressed as $\sum_{j \in N_i} \pi_{ij} f_{a(j,-j)}(t)$ and $\sum_{j \in N_i} \pi_{ij} f_{y(j,-j)}(t)$, where $\pi_{ij} > 0$ denotes the influence coefficient of vehicle j on vehicle i .

The dynamics of the virtual leader vehicle (vehicle 0) are also governed as:

$$\begin{cases} \dot{p}_0(t) = v_0(t) \\ \dot{v}_0(t) = a_0(t) \\ \dot{a}_0(t) = K_0(t) + \frac{u_0(t)}{\tau_0} \\ y_0(t) = p_0(t) \end{cases} \quad (2)$$

where $p_0(t)$, $v_0(t)$, $a_0(t)$, $u_0(t)$, $y_0(t)$, $K_0(t)$, $\tau_0 \in \mathbb{R}$ denote the position, velocity, acceleration, control input, system output, unmodeled dynamics, and engine time lag of the virtual leader

vehicle, respectively. It is assumed that the state information is fully known and that no malicious FDI attacks are present. In a vehicular platoon system, each vehicle is required to maintain a security constraint id while matching the velocity and acceleration of the virtual leader vehicle. Consequently, the tracking error of vehicle i can be expressed as $e_i = [y_i \ v_i \ a_i]^T - [y_0 \ v_0 \ a_0]^T$.

In this study, the design of the malicious FDI attacks exhibit a certain level of intelligence and adaptability. This is reflected in two key aspects. (i) The malicious FDI attacks on vehicle i dynamically adjusts based on the error states of both the vehicle itself and its neighboring vehicles. This enables the attack to achieve a flexible and stealthy impact. The specific formulation is given by:

$$\begin{aligned} f_{a(i,-i)} &= S_i \begin{bmatrix} e_i^T & e_{-i}^T \end{bmatrix}^T \\ f_{y(i,-i)} &= \Upsilon_i \begin{bmatrix} e_i^T & e_{-i}^T \end{bmatrix}^T \end{aligned} \quad (3)$$

where S_i and Υ_i denote time-varying attack gain matrices that modulate the attack signals based on the tracking error, and the term e_{-i} denotes the tracking error of the neighboring vehicles, which is defined in a manner similar to e_i . The distinction lies in that e_i corresponds to the current vehicle i , whereas e_{-i} corresponds to the neighbors of vehicle i . Similarly, the tracking errors of vehicle j and its neighbors, denoted by e_j and e_{-j} , are defined in the same way. (ii) In terms of its activation timing, the attack is triggered when significant fluctuations occur in the state of vehicle i .

Therefore, the error dynamics of the ICVPS can be established as follows:

$$\dot{e}_i = q_i(e_i) + m_i u_i + f_i(e_i, e_{-i}) + \sum_{j \in N_i} \pi_{ij} f_j(e_j, e_{-j}) \quad (4)$$

where $q_i(e_i) = [v_i - v_0 \ a_i - a_0 \ K_i - K_0 - u_0 \ \tau_0^{-1} + w_i]^T$ is the nonlinear component, $m_i = [0 \ 0 \ \tau_i^{-1}]^T$ is the control input matrix and $f_i(e_i, e_{-i}) = [\dot{f}_{y(i,-i)} \ 0 \ f_{a(i,-i)}]^T$ is the attack term.

Define the augmented error vector as $\bar{e}_i \triangleq [e_i^T \ e_{-i}^T]^T$. The network attack can be denoted as $f_i(\bar{e}_i)$ and parameterized as $f_i(\zeta_i, \bar{e}_i)$, where $\zeta_i \in \Omega_i$ is the attack parameter belonging to the parameter space Ω_i , and $\zeta_i = 0$, $f_i \equiv 0$ is denoted as attack-free scenario. Since vehicle i is directly affected by both its own state and that of its neighbors, the pairs $(i, -i)$ and $(j, -j)$ commonly appear together in the formulation.

Remark 2.1: To prevent collisions between vehicles, vehicle i must maintain a direct security distance constraint d with its neighboring vehicles, namely the preceding vehicle $i - 1$ and the following vehicle $i + 1$. In this study, the coupling effects induced by these security constraints are incorporated into $u_i(t)$. In the dynamics (1), $f_{a(j,-j)}(t)$ and $f_{y(j,-j)}(t)$ denote the composite attacks targeting vehicle j , which are then propagated to vehicle i through the communication network. Since the attack signals are transmitted over the same communication channel, the coefficients in the control input and sensor output channels exhibit consistency.

Remark 2.2: In this study, the assumption that an attacker can fully access real-time vehicle state information may indeed

be overly strong. Here, it is adopted as a worst-case scenario to evaluate the upper bound of the system's vulnerability, thereby ensuring that the system is robust enough to withstand even the most powerful potential adversaries. And each vehicle functions both as a sensor carrier and as the execution layer for control commands. Onboard sensors (e.g., cameras, LiDAR, radar) may be tampered with or spoofed, allowing attackers to obtain or forge real-time state information. In addition, since the platoon relies on V2V and V2I communications to exchange state data and commands, cyberattacks on these channels could enable adversaries to access, inject, or alter real-time information and control messages.

B. Control Objectives

The primary control objective of this study is to design a hierarchical game-based OSC framework, which consists of an inner-layer zero-sum game and an outer-layer graph game. In the inner layer, a dynamic adversarial model is established between the defender (controller) and the attacker (FDI attacks), while in the outer layer, a global Nash equilibrium of the ICVPS is achieved. By implementing the OSC to optimize both local and global performance. Specifically, the objective is to achieve the following tasks:

- i) The tracking stability of the vehicular platoon systems is ensured, meaning that vehicles maintain a uniform inter-vehicle distance with a permissible security margin of αd , while their velocity and acceleration remain consistent with those of the virtual leader vehicle, as expressed by:

$$\begin{aligned} \lim_{t \rightarrow \infty} |y_i - y_0 + id| &\leq \alpha d \\ \lim_{t \rightarrow \infty} |v_i(t) - v_0(t)| &= 0 \\ \lim_{t \rightarrow \infty} |a_i(t) - a_0(t)| &= 0 \end{aligned} \quad (5)$$

where α is the security parameter.

- ii) The hierarchical game-based OSC framework achieves simultaneous optimization of both local vehicle performance and global platoon performance. It ensures that each vehicle achieves local optimality while maintaining the optimal platoon formation. Furthermore, the proposed framework ensures the precision of cooperative control and asymptotic stability of the ICVPS under malicious FDI attacks and disturbances.

Assumption 2.1: The velocity $v_i(t)$ and acceleration $a_i(t)$ are continuously differentiable functions. The dynamic functions $K_i(t)$ and disturbances $w_i(t)$ are continuously varying and differentiable. The function $q_i(e_i)$ satisfies zero initial conditions and adheres to the Lipschitz condition.

Assumption 2.2: The FDI attacks function satisfies the zero-origin condition $f_i(\zeta_i, 0) = 0$ and the Lipschitz continuity condition $\|f_i(\zeta_i, \bar{e}_i)\| \leq l_i(\zeta_i) \|\bar{e}_i\|$, where $l_i(\zeta_i)$ is the Lipschitz constant that depends on the attack parameter ζ_i .

Lemma 2.1 ([26]): Consider an input-to-state stable (ISS) system composed of m subsystems, where the dynamics of each subsystem are given by $\dot{x}_i = q_i(x)$, $i \in \mathcal{M}$, where $x_i \in \mathbb{R}^{n_i}$, $x = [x_1^T, \dots, x_m^T]^T$, and $q_i(x_i)$ is a continuous function

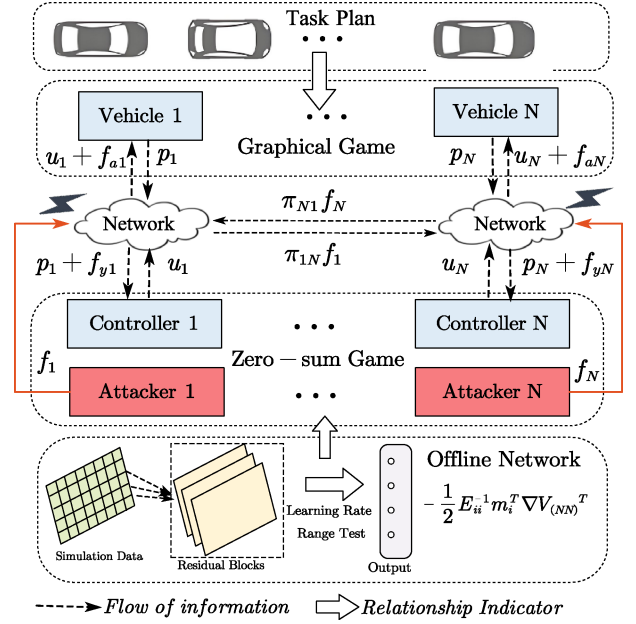


Fig. 2. Hierarchical game-based OSC of ICVPS.

satisfying the Lipschitz condition. Assume that for each subsystem i , there exist gain functions $\gamma_{ij} \in \mathcal{K} \cup \{0\}$, denoting the interconnection gains from subsystem j to subsystem i for all $j \neq i$ and such that:

$$V_i(x_i) \geq \max_{j \neq i} \gamma_{ij}(V_j(x_j)) \Rightarrow \nabla V_i(x_i) q_i(x) \leq -\beta_i(V_i(x_i)) \quad (6)$$

where β_i is a continuous positive definite function.

Furthermore, if for all indices $i_w \in \mathcal{M}$ ($i_w \neq i_{w'}$ for $w \neq w'$), the following condition holds:

$$\gamma_{i_1 i_2} \gamma_{i_2 i_3} \cdots \gamma_{i_w i_1} < 1 \quad (7)$$

then the overall system is asymptotically stable.

III. MAIN RESULT

A novel hierarchical game-based OSC framework is proposed in Fig. 2 to address the security control problem of ICVPS under FDI attacks. In the inner layer, a zero-sum game is formulated between the security controller and the malicious FDI attacker to ensure vehicle-level stability. In the outer layer, a graph game-based optimal control strategy is employed to achieve the Nash equilibrium of the ICVPS, thereby ensuring optimal cooperative precision of the entire ICVPS. To obtain the optimal control strategy efficiently, an OLRNN with improved cosine annealing is designed, which enhances computational efficiency and reduces the complexity of real-world implementation.

A. Zero-Sum Game-Based Security Control in Inner Layer

For the error dynamics (4), both the controller and the attacker can be conceptualized as possessing intelligent, each striving to maximize its own objectives. However, their interests are inherently adversarial. While the controller aims to stabilize the vehicle, the attacker seeks to destabilize it. Consequently, the

control problem for each vehicle can be formulated as a zero-sum differential game. Based on this, the performance function for vehicle i is defined as follows:

$$\begin{aligned} \mathcal{J}_i(\bar{e}_i, u_i, u_j, f_i, f_j) \\ = \int_0^\infty (\bar{e}_i^T Q_i \bar{e}_i + u_i^T E_{ii} u_i + \sum_{j \in N_i} u_j^T E_{ij} u_j \\ - \theta_i f_i^T C_{ii} f_i - \sum_{j \in N_i} \theta_j f_j^T C_{ij} f_j) dt \end{aligned} \quad (8)$$

where Q_i is a positive definite matrix, E_{ii} and C_{ii} are both symmetric positive definite matrices, and C_{ij} is symmetric positive semi-definite. f_i and f_j are abbreviations for $f_i(\bar{e}_i)$ and $f_j(\bar{e}_j)$. The weighting coefficient $\theta_i > 0$. Therefore, the performance index comprehensively accounts for tracking errors, controller energy consumption, and the influence of FDI attacks. It is assumed that the performance function is bounded and continuously differentiable, and the equilibrium point for vehicle i occurs at $\bar{e}_i = 0$. For convenience, the performance index hereafter is denoted by $\mathcal{J}_i(\bar{e}_i)$ or simply $\mathcal{J}_i$. Furthermore, the impact of u_{-i} on u_i is not considered; therefore, it is assumed that u_j always remains at its optimal value u_j^* , and its effect on vehicle i , denoted as E_{ij} , is zero.

The objective of controller is to minimize $\mathcal{J}_i$, whereas FDI attacks aim to maximize $\mathcal{J}_i$. According to the Nash-Pontryagin min-max principle, this interaction constitutes a zero-sum game that admits a unique solution provided that a saddle point exists and satisfies the following equality:

$$\begin{aligned} \min_{u_i} \max_{f_i} \mathcal{J}_i(\bar{e}_i, u_i, u_j^*, f_i, f_j^*) \\ = \max_{f_i} \min_{u_i} \mathcal{J}_i(\bar{e}_i, u_i, u_j^*, f_i, f_j^*) \end{aligned} \quad (9)$$

where u_i^* denotes the optimal control strategy for vehicle i , f_i^* denotes the worst-case attack strategy affecting vehicle i , and f_j^* signifies the worst-case attack strategies from other vehicles.

The Hamiltonian function for vehicle i is defined as follows:

$$\begin{aligned} H_i(\bar{e}_i, \nabla \mathcal{J}_i, u_i, u_j^*, f_i, f_j) \\ = \bar{e}_i^T Q_i \bar{e}_i + u_i^T E_{ii} u_i - \theta_i f_i^T C_{ii} f_i - \sum_{j \in N_i} \theta_j f_j^T C_{ij} f_j \\ + \nabla \mathcal{J}_i^T \left(q_i(\bar{e}_i) + m_i u_i + f_i + \sum_{j \in N_i} \pi_{ij} f_j \right) \end{aligned} \quad (10)$$

where optimal $\mathcal{J}_i^*$ is suit HJI function, $\nabla \mathcal{J}_i$ denotes the gradient of $\mathcal{J}_i$, and $0 = \min_{u_i} \max_{f_i} H_i(\bar{e}_i, \nabla \mathcal{J}_i^*, u_i, u_j^*, f_i, f_j^*)$.

Utilizing the optimality conditions, the optimal control strategy u_i^* and the worst-case fault strategy f_i^* are obtained as:

$$\begin{aligned} u_i^* &= -\frac{1}{2} E_{ii}^{-1} m_i^T \nabla \mathcal{J}_i^* \\ f_i^* &= \frac{1}{2\theta_i} C_{ii}^{-1} \nabla \mathcal{J}_i^* \end{aligned} \quad (11)$$

Consequently, the optimal performance index $\mathcal{J}_i^*$ is expressed as the sum of the cost incurred by vehicle i itself, $\mathcal{J}_{ii}^*$, and the

costs associated with the other terms, $\sum_{j \in N_i} \mathcal{J}_{ij}^*$. The optimal control strategy and the worst-case attack strategy can be expressed as follows:

$$\begin{aligned} u_i^* &= -\frac{1}{2} E_{ii}^{-1} m_i^T \nabla \mathcal{J}_{ii}^* - \frac{1}{2} E_{ii}^{-1} m_i^T \sum_{j \in N_i} \nabla \mathcal{J}_{ij}^* \\ f_i^* &= \frac{1}{2\theta_i} C_{ii}^{-1} \nabla \mathcal{J}_{ii}^* + \frac{1}{2\theta_i} C_{ii}^{-1} \sum_{j \in N_i} \nabla \mathcal{J}_{ij}^* \end{aligned} \quad (12)$$

By substituting u_i^* and f_i^* into the Hamiltonian function and setting it equal to zero, the following is obtained:

$$\begin{aligned} \bar{e}_i^T Q_i \bar{e}_i - \frac{1}{4} (\nabla \mathcal{J}_i^*)^T m_i E_{ii}^{-1} m_i^T \nabla \mathcal{J}_i^* + \frac{1}{4\theta_i} \nabla (\mathcal{J}_i^*)^T C_{ii}^{-1} \nabla \mathcal{J}_i^* \\ + (\nabla \mathcal{J}_i^*)^T q_i(\bar{e}_i) - \frac{1}{2} (\nabla \mathcal{J}_i^*)^T m_i E_{ii}^{-1} m_i^T \nabla \mathcal{J}_i^* \\ + \frac{1}{2\theta_i} (\nabla \mathcal{J}_i^*)^T C_{ii}^{-1} \nabla \mathcal{J}_i^* - \sum_{j \in N_i} \frac{1}{2\theta_j} (\nabla \mathcal{J}_i^*)^T \pi_{ij} C_{jj}^{-1} \nabla \mathcal{J}_j^* \\ - \sum_{j \in N_i} \frac{1}{4\theta_j} (\nabla \mathcal{J}_j^*)^T C_{jj}^{-1} C_{ij} C_{jj}^{-1} \nabla \mathcal{J}_j^* = 0 \end{aligned} \quad (13)$$

where $\mathcal{J}_i^*(0) = 0$ in the above equation.

Assumption 3.1: There exist positive constants a_i, b_i, c_i , and d_i such that the optimal performance index $\mathcal{J}_i^*(\bar{e}_i)$ satisfies:

$$a_i \|\bar{e}_i\|^2 \leq \mathcal{J}_i^*(\bar{e}_i) \leq b_i \|\bar{e}_i\|^2, c_i \|\bar{e}_i\| \leq \|\nabla \mathcal{J}_i^*(\bar{e}_i)\| \leq d_i \|\bar{e}_i\| \quad (14)$$

where $\nabla \mathcal{J}_i^*(\bar{e}_i) \triangleq \frac{\partial \mathcal{J}_i^*(\bar{e}_i)}{\partial \bar{e}_i}$.

Theorem 3.1: Under the Assumptions 2.1, 2.2, and 3.1, the error dynamics of the ICVPS (4) are asymptotically stable under the designed optimal security controller (11) and the following conditions.

1) For each vehicle i , the parameter θ_i satisfies:

$$\theta_i \geq \theta_{i,\min} \triangleq \frac{d_i^2 \|C_{ii}^{-1}\|}{4\lambda_{\min}(Q_i) + \frac{c_i^2}{4} \bar{m}_i \lambda_{\min}(E_{ii}^{-1}) \bar{m}_i} \quad (15)$$

2) For any closed-loop path $i_1 \rightarrow i_2 \rightarrow \dots \rightarrow i_w \rightarrow i_1$ in the ICVPS the gains satisfy:

$$\gamma_{i_1 i_2} \gamma_{i_2 i_3} \dots \gamma_{i_w i_1} < 1 \quad (16)$$

where the gain from V_j to V_i is defined as:

$$\gamma_{ij} \triangleq \frac{h_i \bar{n}_i z_{ij}}{a_j k_i}, z_{ij} \triangleq \frac{d_j^2}{4\theta_j} \|C_{jj}^{-1} C_{ij} C_{jj}^{-1}\| \quad (17)$$

Proof: Consider the Lyapunov function defined as:

$$V_i(\bar{e}_i) = \mathcal{J}_i^*(\bar{e}_i) \quad (18)$$

where $\mathcal{J}_i^*(\bar{e}_i)$ is the optimal value function of the performance index. The derivative of V_i is given by:

$$\dot{V}_i = (\nabla \mathcal{J}_i^*)^T \left(q_i(\bar{e}_i) + m_i u_i^* + f_i^* + \sum_{j \in N_i} \pi_{ij} f_j^* \right) \quad (19)$$

By employing the (10), the following is obtained:

$$\begin{aligned}
(\nabla \mathcal{J}_i^*)^T q_i(\bar{e}_i) &= -\bar{e}_i^T Q_i \bar{e}_i + \frac{1}{4} (\nabla \mathcal{J}_i^*)^T m_i E_{ii}^{-1} m_i^T \nabla \mathcal{J}_i^* \\
&\quad - \frac{1}{4\theta_i} (\nabla \mathcal{J}_i^*)^T C_{ii}^{-1} \nabla \mathcal{J}_i^* \\
&\quad + \frac{1}{2\theta_i} (\nabla \mathcal{J}_i^*)^T C_{ii}^{-1} \nabla \mathcal{J}_i^* \\
&\quad + \sum_{j \in N_i} \frac{1}{4\theta_j} (\nabla \mathcal{J}_j^*)^T C_{jj}^{-1} C_{ij} C_{jj}^{-1} \nabla \mathcal{J}_j^* \\
&\quad + \frac{1}{2} (\nabla \mathcal{J}_i^*)^T m_i E_{ii}^{-1} m_i^T \nabla \mathcal{J}_i^* \\
&\quad + \sum_{j \in N_i} \frac{1}{2\theta_j} (\nabla \mathcal{J}_i^*)^T \pi_{ij} C_{jj}^{-1} \nabla \mathcal{J}_j^* \quad (20)
\end{aligned}$$

Substituting the optimal strategies u_i^* and f_i^* into the Hamiltonian function (10) and setting it equal to zero, the following is obtained:

$$\begin{aligned}
\dot{V}_i &= (\nabla \mathcal{J}_i^*)^T q_i(\bar{e}_i) + (\nabla \mathcal{J}_i^*)^T m_i u_i^* + (\nabla \mathcal{J}_i^*)^T f_i^* \\
&\quad + (\nabla \mathcal{J}_i^*)^T \left(\sum_{j \in N_i} \pi_{ij} f_j^* \right) \\
&= -\bar{e}_i^T Q_i \bar{e}_i + \frac{1}{4} (\nabla \mathcal{J}_i^*)^T m_i E_{ii}^{-1} m_i^T \nabla \mathcal{J}_i^* \\
&\quad + \sum_{j \in N_i} \frac{1}{4\theta_j} (\nabla \mathcal{J}_j^*)^T C_{jj}^{-1} C_{ij} C_{jj}^{-1} \nabla \mathcal{J}_j^* \\
&\quad - \frac{1}{4\theta_i} (\nabla \mathcal{J}_i^*)^T C_{ii}^{-1} \nabla \mathcal{J}_i^* \quad (21)
\end{aligned}$$

By scaling the above equation, the following is obtained:

$$\begin{aligned}
\dot{V}_i &\leq -\lambda_{\min}(Q_i) \|\bar{e}_i\|^2 - \frac{c_i^2}{4} \bar{m}_i \lambda_{\min}(E_{ii}^{-1}) \bar{m}_i \|\bar{e}_i\|^2 \\
&\quad + \frac{d_i^2}{4\theta_i} \|C_{ii}^{-1}\| \|\bar{e}_i\|^2 + \sum_{j \in N_i} \frac{d_j^2}{4\theta_j} \|C_{jj}^{-1} C_{ij} C_{jj}^{-1}\| \|\bar{e}_j\|^2 \\
&= -\left(\lambda_{\min}(Q_i) + \frac{c_i^2}{4} \bar{m}_i \lambda_{\min}(E_{ii}^{-1}) \bar{m}_i - \frac{d_i^2}{4\theta_i} \|C_{ii}^{-1}\| \right) \|\bar{e}_i\|^2 \\
&\quad + \sum_{j \in N_i} \frac{d_j^2}{4\theta_j} \|C_{jj}^{-1} C_{ij} C_{jj}^{-1}\| \|\bar{e}_j\|^2 \quad (22)
\end{aligned}$$

This inequality can be further expressed as:

$$\dot{V}_i \leq -\frac{k_i}{h_i} V_i + \frac{\sum_{j \in N_i} z_{ij}}{a_j} V_j \quad (23)$$

where $k_i \triangleq \lambda_{\min}(\tilde{Q}_i) + \frac{c_i^2}{4} \bar{m}_i \lambda_{\min}(E_{ii}^{-1}) \bar{m}_i - \frac{d_i^2}{4\theta_i} \|C_{ii}^{-1}\|$ and $z_{ij} \triangleq \frac{d_j^2}{4\theta_j} \|C_{jj}^{-1} C_{ij} C_{jj}^{-1}\|$.

To ensure that $\dot{V}_i$ is negative definite, the following conditions must be satisfied:

- 1) $k_i > 0$, which implies that $\theta_i \geq \theta_{i,\min}$.

- 2) The inequality $\frac{k_i}{h_i} V_i \geq \frac{\sum_{j \in N_i} z_{ij}}{a_j} V_j$ holds, thus leading to $V_i \geq \frac{h_i \sum_{j \in N_i} z_{ij}}{a_j k_i} V_j$.

Furthermore, it can be concluded that:

$$V_i \geq \max_{j \in N_i} \left(\frac{h_i \bar{n}_i z_{ij}}{a_j k_i} V_j \right) \quad (24)$$

where $\bar{n}_i$ is the number of neighbors of vehicle i .

Specifically, since:

$$\theta_i \geq \theta_{i,\min} \Rightarrow l_i(\zeta_i) \leq l_{i,\max}(\zeta_i) \triangleq \frac{1}{2\theta_{i,\min}} \|C_{ii}\|^{-1} d_i \quad (25)$$

and

$$V_i \geq \max_{j \in N_i} (\gamma_{ij} V_j) \Rightarrow \dot{V}_i \leq -\Delta_i \|e_i\|^2 \quad (26)$$

where $\Delta_i = \frac{1}{b_i} - \frac{1}{\gamma_{ij}}$. The proof is complete.

Remark 3.1: To satisfy condition 1 of Theorem 3.1, the attack parameters must remain below the worst-case thresholds, i.e., $l_i(\zeta_i) \leq l_{i,\max}(\zeta_i)$. In OSC studies, the assumption of an upper bound on attacks is commonly adopted and aligns well with practical scenarios [12], [14], [15]. For condition 2, it is necessary to ensure that each loop within the ICVPS satisfies the specified requirement. However, since the topology of the ICVPS is modeled as a single-loop system in this study, and each vehicle is assumed to be subject to FDI attacks, all vehicle controllers must be appropriately adjusted to meet this condition. Given that these conditions are satisfied, the game-based OSC approach enables controllers to exhibit intelligent security control capabilities, allowing them to formulate optimal strategies to minimize the cost function and adapt in real time to the occurrence of malicious FDI attacks.

B. Graph Game-Based Optimal Control in Outer Layer

This section establishes a graph game model for the ICVPS by applying graph theory to describe the interactions among vehicles within the system and investigates its Nash equilibrium. Each vehicle $i \in \mathcal{N}$ is considered a player in the graph game, leading to the following definition of the graph game.

Definition 3.1: If $\pi_{ij} \neq 0$, $C_{ij} \neq 0$, then the graph game formed by the ICVPS is well-defined.

Definition 3.2: For n vehicles, if there exist n -tuple of policies $\{(u_1^*, f_1^*), (u_2^*, f_2^*), \dots, (u_n^*, f_n^*)\}$ such that:

$$\begin{cases} \mathcal{J}_i(u_i^*, u_j^*, f_i^*, f_j^*) \leq \mathcal{J}_i(u_i^*, u_j^*, f_i^*, f_j^*) \\ \mathcal{J}_i(u_i, u_j^*, f_i^*, f_j^*) \geq \mathcal{J}_i(u_i^*, u_j^*, f_i^*, f_j^*) \end{cases} \quad (27)$$

then the strategy set constitutes a Nash equilibrium.

Theorem 3.2: For the ICVPS, the strategy set $\{(u_1^*, f_1^*), (u_2^*, f_2^*), \dots, (u_n^*, f_n^*)\}$ forms a Nash equilibrium. Under this equilibrium, the performance index for each vehicle attains its optimal value, explicitly given by:

$$\mathcal{J}_i^*(\bar{e}_i) = V_i(\bar{e}_i), \quad \forall i \in \mathcal{N} \quad (28)$$

This result holds provided that Assumptions 2.1 and 2.2 are satisfied, $V_i(\bar{e}_i) > 0$ is a positive solution to the HJI equation,

and the optimal control strategy u_i^* and worst-case attack strategy f_i^* are designed in (12).

Proof: For the functions V_i , $i \in \mathcal{N}$, the performance index can be rewritten as:

$$\begin{aligned} \mathcal{J}_i(\bar{e}_i, u_i, u_j, f_i, f_j) &= \int_0^\infty (\bar{e}_i^T Q_i \bar{e}_i + u_i^T E_{ii} u_i + \sum_{j \in N_i} u_j^T E_{ij} u_j \\ &\quad - \theta_i f_i^T C_{ii} f_i - \sum_{j \in N_i} \theta_j f_j^T C_{ij} f_j) dt \\ &\quad + \int_0^\infty \nabla V_i^T (q_i + m_i u_i + f_i + \sum_{j \in N_i} \pi_{ij} f_j) dt \\ &\quad + V_i(\bar{e}_i) - V_i(\bar{e}_i(\infty)) \end{aligned} \quad (29)$$

where $V_i(\bar{e}_i(\infty)) = V_i(\lim_{t \rightarrow \infty} \bar{e}_i(t))$.

The following is obtained by applying the HJI equation:

$$\begin{aligned} \bar{e}_i^T Q_i \bar{e}_i + \nabla V_i^T q_i &= (u_i^*)^T E_{ii} u_i^* - \theta_i (f_i^*)^T C_{ii} f_i^* + \sum_{j \in N_i} \theta_j (f_j^*)^T C_{ij} f_j^* \\ &\quad - \nabla V_i^T \sum_{j \in N_i} \pi_{ij} f_j^* \end{aligned} \quad (30)$$

By substituting this into the performance index, the following equation is obtained:

$$\begin{aligned} \mathcal{J}_i &= V_i(\bar{e}_i) - V_i(\bar{e}_i(\infty)) + \int_0^\infty \left[\bar{e}_i^T Q_i \bar{e}_i + u_i^T E_{ii} u_i \right. \\ &\quad - \theta_i f_i^T C_{ii} f_i - \sum_{j \in N_i} \theta_j f_j^T C_{ij} f_j + \nabla V_i^T m_i u_i + \nabla V_i^T f_i \\ &\quad + \nabla V_i^T \sum_{j \in N_i} \pi_{ij} f_j + (u_i^*)^T E_{ii} u_i^* - \theta_i (f_i^*)^T C_{ii} f_i^* \\ &\quad \left. + \sum_{j \in N_i} \theta_j (f_j^*)^T C_{ij} f_j^* - \nabla V_i^T \sum_{j \in N_i} \pi_{ij} f_j^* \right] dt \end{aligned} \quad (31)$$

By completing the squares, the following equation is obtained:

$$\begin{aligned} \mathcal{J}_i(\bar{e}_i, u_i, u_j, f_i, f_j) &= V_i(\bar{e}_i) - V_i(\bar{e}_i(\infty)) + \int_0^\infty \left[(u_i - u_i^*)^T E_{ii} (u_i - u_i^*) \right. \\ &\quad - \theta_i (f_i - f_i^*)^T C_{ii} (f_i - f_i^*) - 2 \sum_{j \in N_i} \theta_j (f_j^*)^T C_{ij} (f_j - f_j^*) \\ &\quad - \sum_{j \in N_i} \theta_j (f_j - f_j^*)^T C_{ij} (f_j - f_j^*) \\ &\quad \left. + \nabla V_i^T \sum_{j \in N_i} \pi_{ij} (f_j - f_j^*) \right] dt \end{aligned} \quad (32)$$

Based on Theorem 3.1, the ICVPS are asymptotically stable. Therefore, $V_i(\bar{e}_i(\infty)) = 0$. Assuming that the other vehicle j

employs the optimal strategy, i.e., $f_j = f_j^*, \forall j \in N_i$, the following equation is derived:

$$\begin{aligned} \mathcal{J}_i(\bar{e}_i, u_i, u_j^*, f_i, f_j^*) &= V_i(\bar{e}_i) + \int_0^\infty \left[(u_i - u_i^*)^T E_{ii} (u_i - u_i^*) \right. \\ &\quad \left. - \theta_i (f_i - f_i^*)^T C_{ii} (f_i - f_i^*) \right] dt \end{aligned} \quad (33)$$

To validate Definition 3.2, a systematic analysis of the following three scenarios is conducted:

Scenario 1: The control input deviates from the optimal strategy, i.e., $u_i \neq u_i^*$, while the attack strategy remains optimal, i.e., $f_i = f_i^*$. In this scenario, the performance index $\mathcal{J}_i$ is expressed as:

$$\mathcal{J}_i^1 = V_i(\bar{e}_i) + \int_0^\infty \left((u_i - u_i^*)^T E_{ii} (u_i - u_i^*) \right) dt \quad (34)$$

Since the integral term $(u_i - u_i^*)^T E_{ii} (u_i - u_i^*)$ is always positive, it follows that $\mathcal{J}_i > V_i(\bar{e}_i)$.

Scenario 2: Both the control input and attack strategy adopt the optimal strategies, i.e., $u_i = u_i^*$ and $f_i = f_i^*$. In this scenario, the performance index simplifies to:

$$\mathcal{J}_i^2 = V_i(\bar{e}_i) \quad (35)$$

indicating that $\mathcal{J}_i$ attains its optimal value.

Scenario 3: The control input adopts the optimal strategy, i.e., $u_i = u_i^*$, but the attack strategy deviates from the optimal, i.e., $f_i \neq f_i^*$.

Here, the performance index is expressed as:

$$\mathcal{J}_i^3 = V_i(\bar{e}_i) - \int_0^\infty \theta_i (f_i - f_i^*)^T C_{ii} (f_i - f_i^*) dt \quad (36)$$

Since the integral term $-\theta_i (f_i - f_i^*)^T C_{ii} (f_i - f_i^*)$ is negative, it follows that $\mathcal{J}_i \leq V_i(\bar{e}_i)$.

Based on the above analysis, the following inequality holds:

$$\mathcal{J}_i^{(3)} \leq \mathcal{J}_i^{(2)} \leq \mathcal{J}_i^{(1)} \quad (37)$$

This relationship indicates that employing the optimal control strategy u_i^* and the optimal attack strategy f_i^* enables the performance index to attain its optimal value $V_i(\bar{e}_i)$. Any deviation from these strategies can either degrade the performance index or fail to improve it, thus satisfying the requirements of Definition 3.2.

Furthermore, assuming that all other vehicles adopt their respective optimal strategies, if vehicle i attempts to unilaterally modify its optimal control input u_i^* or attack strategy f_i^* , its performance index $\mathcal{J}_i$ can either increase or decrease, resulting in no performance improvement. Consequently, vehicle i cannot enhance its performance by deviating from its optimal strategies. The proof is complete.

Remark 3.2: Theorem 3.2 demonstrates that the optimal performance of vehicle i depends on both its own strategy (u_i^*, f_i^*) and the strategies of other vehicles. This equilibrium condition ensures strict Nash optimality—no unilateral deviation can improve individual performance without reducing the overall coordination efficiency of the ICVPS. The proposed graph game

framework does not result in performance trade-offs between local and global objectives [17], [23], [27]. Specifically, the hierarchical game-based OSC framework presented in this study achieves global optimality for the ICVPS while ensuring that all vehicles simultaneously meet their respective local optimal performance criteria. This holds true even in the presence of malicious FDI attacks and external disturbances, optimizing both the system's coordination performance and the stability of individual vehicles.

Remark 3.3: The adoption of a hierarchical game theory framework significantly reduces the computational complexity of large-scale systems. The proposed method decomposes the global problem into local subproblems through a layered design: the upper layer is responsible for platoon formation planning or reference strategy generation, while the lower layer ensures that each vehicle only addresses adversarial interactions within its local neighborhood. As a result, each subproblem involves a limited number of state dimensions, and the complexity of solving or approximating the HJI equation does not grow exponentially with n , but rather remains relatively constant. Additionally, the control decisions for each vehicle can be computed in parallel and independently, requiring only information from itself and its neighboring vehicles.

C. Offline-Learning Residual Neural Network

In this section, a method for determining the optimal control strategy by learning the value function through residual neural networks is presented. To this end, a neural network $V_{NN}(x; \theta_{NN})$, parameterized by θ_{NN} , is designed. The network architecture comprises an input layer, multiple hidden layers with residual connections, and an output layer.

Each residual block implements the following transformation:

$$h_{out} = h_{in} + \sigma(Wh_{in} + b) \quad (38)$$

where h_{in} and h_{out} denote the input and output of the block, W and b are the weights and biases, and σ is the activation function.

A specialized offline training database is developed to address the scenarios under consideration. Specifically, data are collected from simulations of the ICVPS, conducted under identical conditions but in the absence of malicious FDI attacks. This dataset is subsequently generalized and employed as the training database for the OLRNN described in this section. By utilizing the same models and conditions as those in the original scenario, the trained network exhibits high specificity, thereby improving the approximation accuracy of the OLRNN.

To enhance the training effectiveness of the residual network, an improved cosine annealing algorithm is adopted to dynamically adjust the learning rate. During the initial training phase, the maximum learning rate $\eta_{max}(tk)$ is estimated, and the learning rate η_{tk} is increased exponentially as follows:

$$\eta_{tk} = \eta_{start} \times \left(\frac{\eta_{end}}{\eta_{start}} \right)^{\frac{tk}{Tk}} \quad (39)$$

where η_{start} and η_{end} denote the initial and final learning rates, respectively, tk is the current iteration index, and Tk is the total number of iterations. In each iteration, the corresponding loss

$\mathcal{L}_{tk}$ is recorded, and $\eta_{max}(tk)$ is dynamically determined based on these data. By plotting the relationship between $\log(\eta_{tk})$ and $\mathcal{L}_{tk}$, the learning rate at which $\mathcal{L}_{tk}$ starts to rise abruptly is identified as $\eta_{max}(tk)$.

Once $\eta_{max}(tk)$ is obtained, it is incorporated into the improved cosine annealing formula to compute the current iteration's learning rate η_{tk} :

$$\eta_{tk} = \eta_{min} + \frac{1}{2}(\eta_{max}(tk) - \eta_{min}) \left[1 + \cos \left(\frac{Tk_{cur}}{Tk_{max}} \pi \right) \right] \quad (40)$$

where η_{min} is the minimum allowable learning rate, Tk_{cur} is the current iteration within the cosine cycle, and Tk_{max} is the total number of iterations in one cycle.

By dynamically determining $\eta_{max}(tk)$, the learning rate η_{tk} consistently satisfies the stability conditions of gradient descent and adaptively responds to variations, thereby enhancing the optimizer's ability to escape local optima.

Subsequently, the training process minimizes the mean squared residual:

$$\begin{aligned} \mathcal{L}(\theta_{NN}) = \mathbb{E}_{\bar{e}_i \sim \rho(\bar{e}_i)} & \left[\left(L(\bar{e}_i, u^*, f^*) \right. \right. \\ & \left. \left. + \nabla V_{NN}(\bar{e}_i; \theta_{NN})^T \mathbf{D}(\bar{e}_i, u^*, f^*) \right)^2 \right] \end{aligned} \quad (41)$$

where $\rho(\bar{e}_i)$ denotes the distribution of state samples, $\mathbf{D}$ are the dynamics of system, and u^* and f^* depend on $\nabla V_{NN}(\bar{e}_i; \theta_{NN})$.

Remark 3.4: The time complexity of training deep neural networks is proportional to the product of the number of training samples, the size of the network parameters, and the number of training iterations. In the OLRNN framework, the introduction of residual blocks does not increase the computational complexity, ensuring that the time complexity remains at a polynomial level, significantly lower than the direct solution of the HJI equation. The space complexity is primarily determined by the number of model parameters, which remains within the capacity of modern GPUs and memory resources. Since OLRNN training is entirely performed offline and conducted on high-performance cloud computing platforms, the online phase only requires loading the trained network parameters, enabling rapid real-time computation.

Through the design of the OLRNN, the training efficiency of the model can be significantly improved while effectively avoiding entrapment in local optima. Moreover, by adopting an offline training approach, the risks of collisions, instability, and other security issues during actual deployment are substantially reduced. The specific algorithmic implementation steps are provided in Algorithm 1.

IV. SIMULATION

This section validates the hierarchical game-based OSC algorithm using the CARLA simulator. The experiment considers an ICVPS consisting of one virtual leader and four follower vehicles. The nonlinear vehicle dynamics is $K_i = -(a_i + l_i v_i^2 M_i^{-1} + o_i M_i^{-1}) \tau_i^{-1} - 2l_i v_i a_i M_i^{-1}$, where l_i , o_i denote air resistance coefficient and mechanical resistance.

TABLE I
PARAMETERS OF THE VEHICLES IN ICVPS

Vehicle node i	1	2	3	4
L_i (m)	3.5	4.5	5	5
τ_i (s)	0.4	0.85	0.5	0.55
M_i ($\times 10^3$ kg)	1.8	1.85	1.8	1.9

Algorithm 1: Improved Cosine Annealing Learning Rate Scheduler with Range Test.

Require: • *optimizer*, $\eta_{\min}$, η_{start} , η_{end} , N_k , $Tk_{\max}$, Tk , $\mathcal{J}$

- 1: Initialize: $Tk_{\text{cur}} \leftarrow 0$
- 2: **for** epoch = 1 $\rightarrow Tk$ **do**
- 3: **(A) Training Phase:** Perform training, compute loss $\mathcal{J}$
- 4: **(B) Learning Rate Range Test:**
- 5: $\eta \leftarrow \eta_{\text{start}}$, $r \leftarrow \left(\frac{\eta_{\text{end}}}{\eta_{\text{start}}}\right)^{\frac{1}{N_k}}$
- 6: **for** test_iter = 1 $\rightarrow N_k$ **do**
- 7: $\eta \leftarrow \eta \times r$, set optimizer's learning rate to η
- 8: Perform training step, compute L_{test} , record L_{test}
- 9: **if** L_{test} increases **then**
- 10: $\eta_{\max} \leftarrow \frac{\eta}{r}$, **break**
- 11: **end if**
- 12: **end for**
- 13: **(C) Update LR via Cosine Annealing:**
- 14: $\eta \leftarrow \eta_{\min} + \frac{1}{2}(\eta_{\max} - \eta_{\min}) \left[1 + \cos\left(\frac{\pi Tk_{\text{cur}}}{Tk_{\max}}\right)\right]$
- 15: **(D) Update Optimizer's LR:** Set each parameter group's LR to η
- 16: **(E) Increment Cycle:**
- 17: $Tk_{\text{cur}} \leftarrow (Tk_{\text{cur}} + 1) \bmod Tk_{\max}$
- 18: **(F) Record State:** Record(epoch, η , $\mathcal{J}$, $\eta_{\max}$)
- 19: **end for**

The external disturbance is set as $w_i(t) = 0.1 \sin(t)$, and the control input is bounded by $\|u_i(t)\| \leq 150 \text{ N} \cdot \text{m}$. The longitudinal parameters are chosen as $l_i = 0.34$, $o_i = 0.1 \text{ N}$, desired inter-vehicle distance $d = 10 \text{ m}$, and security factor $\alpha = 0.2$. Other relevant parameters are listed in Table I [28].

The acceleration of the virtual leader vehicle $i = 0$ is defined as:

$$a_0(t) = \begin{cases} 0.48t \text{ m/s}^2 & 1 \text{ s} \leq t \leq 5 \text{ s} \\ 2.4 \text{ m/s}^2 & 5 \text{ s} \leq t \leq 10 \text{ s} \\ -0.48t + 7.2 \text{ m/s}^2 & 10 \text{ s} \leq t \leq 15 \text{ s} \\ 0 \text{ m/s}^2 & \text{otherwise} \end{cases}$$

Considering the malicious FDI attacks in the controller channel $f_{a(i,-i)}$ and in the sensor channel $f_{y(i,-i)}$, the fault functions are $S_i(t) = 0.2 \cos(t) I_{1 \times 6}$ and $\Upsilon_i(t) = 0.1 \sin(2t) I_{1 \times 6}$. Since malicious FDI attacks are more intelligent and targeted compared to traditional FDI attacks, these attacks typically occur during critical moments of velocity variation in the ICVPS. In Fig. 3, two 2-second-long malicious FDI attacks occur at key moments: during vehicle deceleration, $9 \text{ s} \leq t \leq 11 \text{ s}$, and during the transition to constant velocity, $14 \text{ s} \leq t \leq 16 \text{ s}$. In this study, malicious FDI attacks are considered capable of obtaining

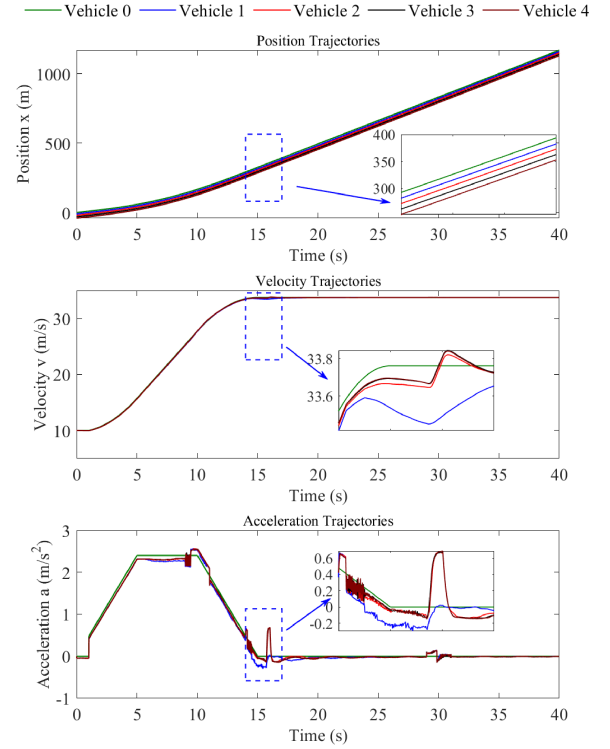


Fig. 3. Trajectories of leader and follower vehicles.

the system's current state information. Therefore, these attacks are deemed to occur during phases when the system's stability is more vulnerable, particularly when the state of the system undergoes significant changes. In the current ICVPS, this vulnerability manifests as attacks typically occurring at key moments of velocity variation. The duration of these attacks is set to 2 seconds per cycle at velocity change points, which is deemed sufficient to cause significant disruption to the system, taking into account the limited resources available for the attack. Additionally, a 2-second-long conventional FDI attack is implemented during the period of $29 \text{ s} \leq t \leq 31 \text{ s}$ as a comparative case to demonstrate the differing impacts between variable-velocity and constant-velocity phases. This comparison effectively illustrates the distinction between malicious FDI attacks and conventional FDI attacks. Notably, the interference caused by the first two malicious FDI attacks (during the deceleration and transition phases) is significantly more pronounced than that in the third constant-velocity phase. This observation validates the effectiveness of the proposed algorithm in mitigating malicious FDI attacks, particularly during critical velocity variation periods that demand higher control precision in ICVPS.

Figs. 3 to 6 clearly present the comparative simulation results between the proposed method and the network attack defense scheme based on the adaptive control law with a fused state estimation compensator [28]. Using the same scenario and model parameters, Figs. 3 and 4 both show the state tracking performance of the ICVPS. It is evident that, under attack, the proposed method demonstrates faster convergence and smaller maximum errors compared to the benchmark method. Despite the controller and sensor channels in the platoon system being subjected to malicious FDI attacks at critical velocity change points, the states of all follower vehicles remain synchronized

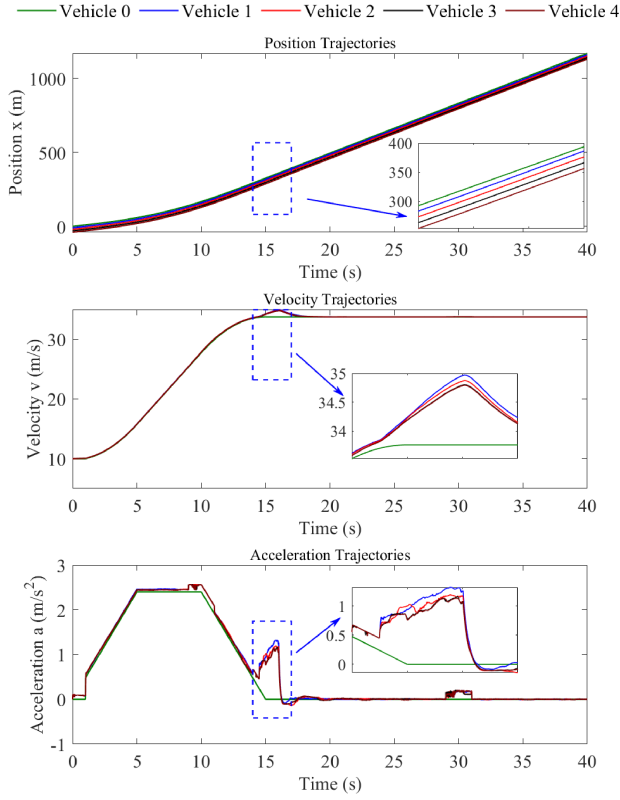


Fig. 4. Trajectories of leader and follower vehicles based on [28].

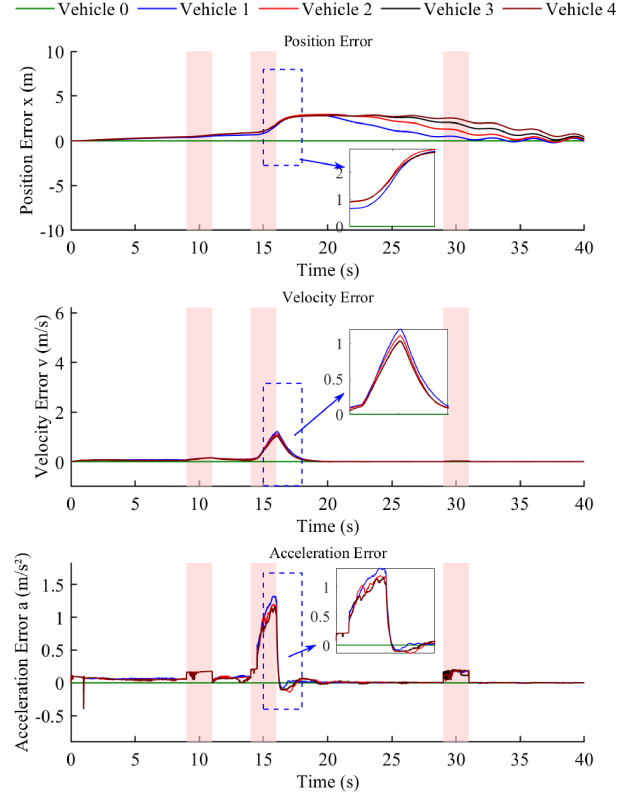


Fig. 6. Tracking errors between follower vehicles based on [28].

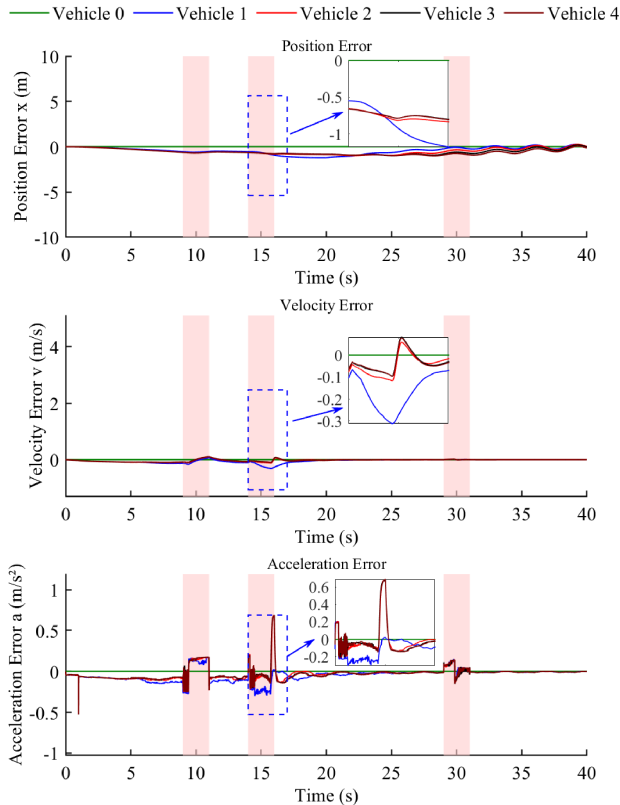


Fig. 5. Tracking errors between follower vehicles.

with the leader vehicle's state. Although the attacked vehicles exhibit significant fluctuations in velocity and acceleration, the proposed controller effectively restores them to their normal state within a short period.

Additionally, Figs. 5 and 6 show the tracking errors of vehicles in the ICVPS. It is clear that, compared to the benchmark method, the proposed method results in smaller peak values and less fluctuation in acceleration, velocity, and position errors. Furthermore, according to the control objective of this study, i.e., $\lim_{t \rightarrow \infty} |y_i - y_0 + id|$, the benchmark method fails to achieve this control objective, while under the proposed controller, the position deviations of all vehicles remain within the predefined safety distance threshold, without exceeding αd . A snapshot from the CARLA simulation in Fig. 7 clearly demonstrates that the hierarchical game-based OSC algorithm enables the vehicles to converge smoothly and quickly to the reference trajectory.

Thus, through the comparative simulations, the effectiveness of the proposed hierarchical game-based OSC algorithm is validated, highlighting its robust control performance in the presence of malicious FDI attacks and external disturbances, as well as its ability to achieve high cooperative control accuracy.

Figs. 8 and 9 demonstrate the performance of the OLRNN through ablation experiments. Specifically, Fig. 8 illustrates the process of estimating the maximum allowable learning rate, highlighting how varying learning rates influence the decline of the experimental loss function. Additionally, as shown in Fig. 9(a) and (b), three distinct approaches are compared: the proposed improved method (denoted by a blue solid line), a constant learning rate baseline (denoted by an orange dashed

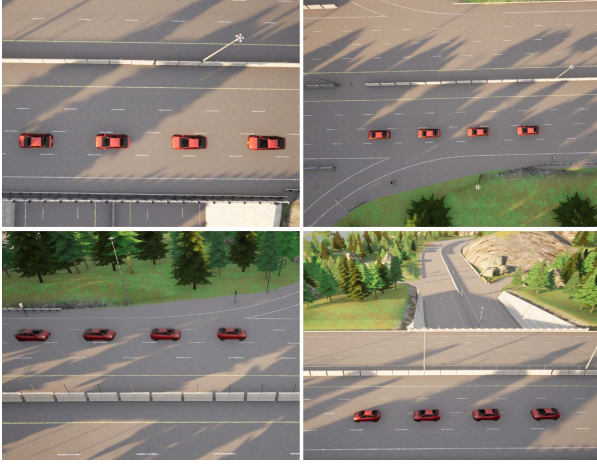


Fig. 7. CARLA simulation results.

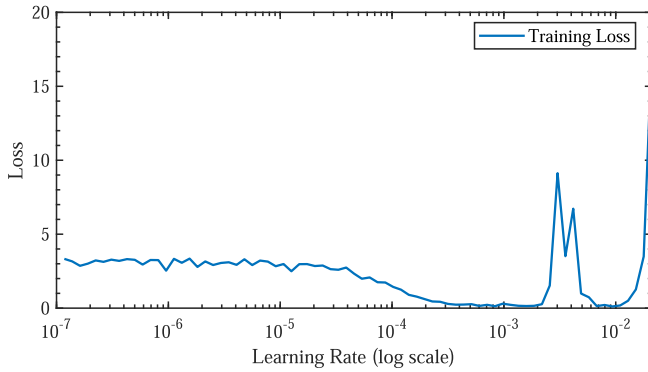


Fig. 8. Learning rate range test.

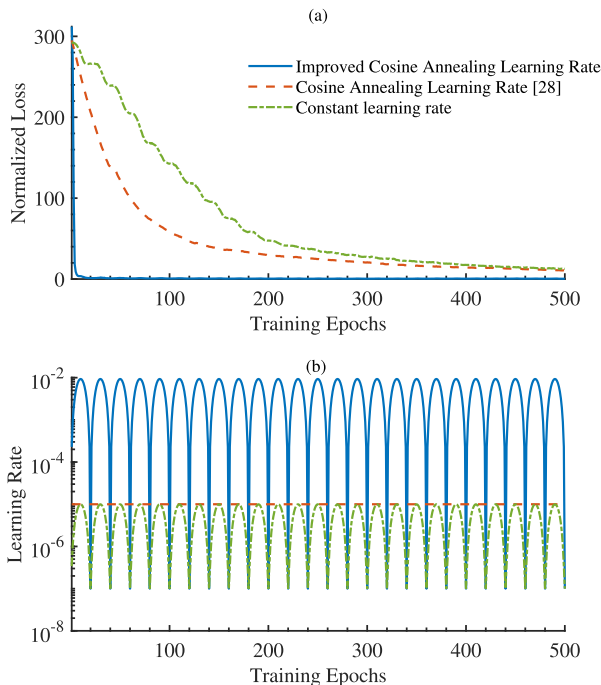


Fig. 9. (a) Loss evolution (b) Learning rate schedule.

line), and conventional cosine annealing (denoted by a green dotted line). The results clearly show that the proposed method outperforms in accelerating loss function reduction, while maintaining sustained effectiveness throughout the subsequent optimization phases compared to the baseline approaches [29]. This indicates that the improved cosine annealing algorithm proposed in this study enables more efficient and stable training of residual networks, enhancing their applicability in real-world control systems.

V. CONCLUSION

This study proposes a hierarchical game-based OSC framework to address the security challenges of ICVPS under malicious FDI attacks and external disturbances. The framework integrates outer-layer graph games and inner-layer zero-sum games, optimizing global coordination performance and local stability while ensuring platoon tracking control. The efficient training of the controller is achieved through the OLRNN. Simulation results demonstrate that the proposed method effectively mitigates malicious FDI attacks, ensuring the security, stability, and collaborative accuracy of the vehicular platoon, showcasing strong robustness and practical application potential. Future research can extend the current framework to address ICVPS subjected to hybrid types of network attacks and system faults. Additionally, due to the limitations of offline learning in continuously improving control performance, a combination of offline learning and online reinforcement learning is expected to significantly enhance the control accuracy, robustness, and attack resistance of ICVPS.

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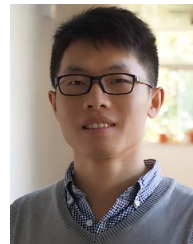


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